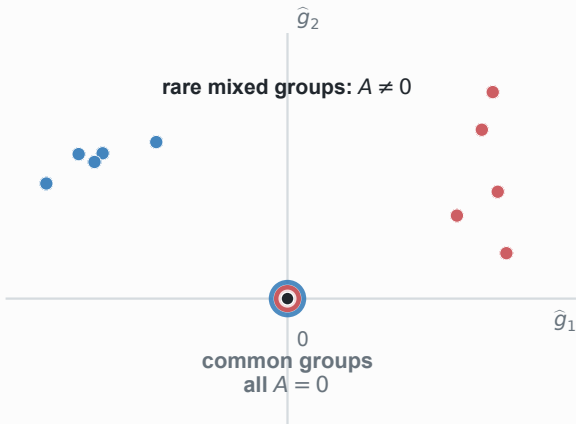


● task i

● task j



$$P_{\text{mix}} = 1 - q^G - (1 - q)^G \\ = G(1 - q) + O((1 - q)^2)$$